

# A Non-Linear Approach to Predict the Salary of NBA Athletes using Machine Learning Technique

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**Abstract—** Every sportsman traded/drafted receives monetary compensation in accordance with their contract. In this study, we propose a nonlinear approach based on performance and other aspects to determine the salary of a basketball player. We estimate the salary based on four regressive models. Whilst predicting we also figure out the important features impacting the salary. Comparatively speaking, random forest outperformed other algorithms. Furthermore, we consider that our findings might benefit discussions between basketball teams and players. This model can also help set a benchmark for salary expectations by the players in accordance.

**Keywords—** Non-Linear Model, Regression machine learning, Salary prediction, Sports analytics, XGBoost, Random Forest

## I. INTRODUCTION

This National Basketball Association (NBA) is a professional sporting league in North America and is the most renowned basketball league in the world. One of the top professional sports leagues in the United States and Canada, the league has 30 teams. The teams are divided into two conferences – Eastern and Western. The teams from each conference (East and West) are ranked based on number of games won or lost. Based on their ranking the top six teams qualify for the playoffs. Teams with ranks 7-10 enter a Play-In Tournament from each conference. And the top two teams advance to the Playoffs of their respective conference. In the end, teams from each conference play against one another to determine who is the best team based on wins. After which a series of seven matches are held amongst the finalists of Western conference and Eastern conference. This decides the ultimate champion team of the particular year or season.

In recent times, NBA's business worth is on a stride with millions of viewers and commercials. Given the athlete's popularity and demand, management's perspective is centered on the decision to select the player. Although modern technology and sports analytics provide great support to pick players. Choosing them has a cost.

The cost can be thought of as salary plus other benefits. One core component based on which a player is chosen by a team is their performance. Other attributes like Age, previous salary, position, fantasy points, etc. can also be utilized to pick a player for the team. Machine learning can be deployed to comprehend patterns that enable analysis and prediction.

Such results can be exploited by the Team manager to negotiate monetary terms with the player.

There exists widespread research by various researchers to predict the salary of an athlete based on several aspects. However, the crucial fact is to check if linearity is viable before predicting the salary. Most of the models attempt to find a relationship based on attributes and salary forgoing linearity or nonlinearity present between the variables. The first model that is often run is multiple linear regression, which does not appear to be a viable model to forecast since there may be a non-linear relationship between the qualities and pay. we focus on nonlinear regressive models to predict pay in this study. according to our findings, the random forest regressor is a superior predictor.

The rest of the paper is organized as follows. The literature review is narrated in the second section. Third section explains the Methodology. Followed by data-set explanation and Machine Learning Algorithms in fourth and fifth section respectively. Evaluation Metrics & Results are explained in sixth section. Lastly the conclusion is stated in seventh section.

## II. LITERATURE REVIEW

Mustafa A.Al-Asadi and Sakir Tasdemir [1] demonstrated how machine learning algorithms may be used to forecast football player worth. Transfer fees that may be paid for a football player on the football market provide a decent indication of a player's market value. Traditionally the market has been valued by football experts. However, the judgments are not accurate. The study makes use of FIFA 20 video game data obtained from sofifa.com. Four regression models—linear regression, multiple linear regression, decision trees, and random forests—that were tested on the entire collection of variables were used to estimate market worth for the players. Eventually, they believe that our findings can considerably assist conversations between football teams and a player's agency. They conclude in their study that these models can be utilized as a starting point to quantitatively determine a player's market value.

Nuoya Li [2] establishes a link between on-court performance, individual traits, and pay in addition to conducting two regression analyses were performed to investigate the factors that influence compensation and overpayment in the first year of a new contract. In the first regression, he observed that a player's wage is impacted by

Fadi Thabtan, Li Zhang et.al., [3] - The first step is to identify the crucial statistics that include the most details about the players' pay. Secondly, we predict the NBA player salaries using the selected statistics. The Least Absolute Shrinkage and Selection Operator (LASSO) was used to select the essential data, and the Random Forest (RF) approach was employed to estimate wages. They demonstrated the need to use non-linear models and algorithms since NBA player statistics and compensation have a clearly non-linear relationship. Additionally, they demonstrated how to quantify a model's predictability as well as the proper method for examining the link between a response and a large number of predictor variables. The RandomForest non-linear algorithm was used to forecast player salaries after the researchers used LASSO variable selection to identify the significant elements (statistics) that are most frequently linked to NBA player's salary salaries. The accuracy of forecasts made about NBA player wages based on the players' on-court performance is acceptable, but not as good as one might like. Researchers argue that important characteristics included in the study, such as popularity, the quantity of spectacle offered, and so on, might significantly improve the accuracy of pay projections.

The method employed in this work is supervised machine learning. Another pointer to keep in mind is that, in Machine Learning we have the actual values of the variable we wish to predict. In such a learning technique the algorithm learns from a part of the dataset to come up with a model (also known as training the model).

Fig. 1. Proposed Framework for NBA Salary Prediction

*A. Scrutiny of variables / attributes to understand the factors that impact player's salary while predicting:*

To study the relativeness and quality of the features with salary (the value to be predicted) Pearson Correlation Coefficient was used. In addition, Feature importance function was used from “scikit-learn” in Python. In general, valuable features tend to be highly correlated with the variable to be predicted, salary in our case. Fig 2 showcases

Fig. 2. Correlations of player's Performance Measures to Salary of Players

Fig. 3. Distribution of Salary of Players



### E. Data Splitting

#### F. Building various models to predict the salary

forests, and XGBoost). These models can be built from the module called `scikit-learn` in Python.

Various measures like Root mean square errors (RMSE), Mean absolute errors (MAE), and the coefficient of determination (R2) was utilized to gauge the accuracy of the models with the testing dataset. However, before gauging the model, hyperparameter tuning was carried out. A parameter used to regulate how the model learns is known as a hyperparameter and was used to regulate the learning process. Tuning them is the process of choosing the optimal hyperparameters with an aim of improving the ability of a model to predict

The dataset used was scrapped from NBA and ESPN's official website as mentioned earlier for the year 2020-21. Stats of previous years were considered in accordance with the salary for next year as shown in Table.1. The reason for this is that a player's estimated salary is determined before the season begins. Moreover, players who did not have physical minutes played in the game of the regular season for the year 2020-21 were not taken into consideration as their performance attributes will not be available.

| Notation | Meaning                       |
|----------|-------------------------------|
| +/-      | Plus-Minus                    |
| 3P%      | 3 Point Field Goal Percentage |
| 3PA      | 3 Point Field Goals Attempted |
| 3PM      | 3 Point Field Goals Made      |
| Age      | Age                           |
| AST      | Assists                       |
| BLK      | Blocks                        |
| DD2      | Double Doubles                |
| DREB     | Defensive Rebounds            |
| FG%      | Field Goal Percentage         |
| FGA      | Field Goals Attempted         |
| FGM      | Field Goals Made              |
| FP       | Fantasy Points                |
| FT%      | Free Throw Percentage         |
| FTA      | Free Throws Attempted         |
| FTM      | Free Throws Made              |
| GP       | Games Played                  |
| L        | Losses                        |
| MIN      | Minutes Played                |
| OREB     | Offensive Rebounds            |
| PF       | Personal Fouls                |
| PTS      | Points                        |
| REB      | Rebounds                      |
| STL      | Steals                        |
| TD3      | Triple Doubles                |
| TOV      | Turnovers                     |
| W        | Wins                          |

The performance characteristics (independent variables) were scaled to unit variance using a scaler function centered on the mean. This was done to get rid of the range (as each variable has a different range) and units. For example Age ranges from 20-40 whereas assist ranges between 0-11. Second, age is represented in years, while minutes played are recorded in minutes. Moreover, the position is a categorical feature that was converted into numeric using Label Encoding.

The ability of performance metrics and other attributes to predict salary has advanced in past decades [11]. Besides, enormous efforts have been invested to analyze the performance attributes of basketball athletes to enable managers to avoid overpaying the player. NBA datasets exhibited effectiveness in predicting player's salaries and other analysis. We considered player statistics of the year 2020-2021 (the previous year) to predict the salary of 2022 as a player's contract is signed during the start of the season.

## V. MACHINE LEARNING ALGORITHMS

To predict the player's salary based on performance attributes and other attributes of a basketball player, various Supervised Learning methods were cast. Because the data contains expected output values, supervised machine learning algorithms are used. All of these strategies try to capture and use the relationship between pay prediction and performance and other parameters to forecast. Since the dataset is numeric in nature, Regression techniques are used. Multiple Linear Regression, Decision Tree Regressor, Random Forest Regressor, and XGBoost Regressor are the models deployed. Few of them are frequently used in other studies for describing players and data mining domains [4]. Additionally, algorithms were compared between nonlinear models (such as random forest, and decision trees) and linear models (such as multiple linear regression). Linear regression in statistics is the establishment of a linear technique to forecasting based on the relationship between a dependent and one independent variable (also known as response and predictor variables).

### A. Multiple Linear Regression

Multiple linear regression is a variant of linear regression that includes more than one independent or explanatory variable. . In simple terms, it is the predictor (independent) and predicted (dependent) variables. The model aims to find the best coefficients (betas) for the variables to predict the outcome as close as possible keeping in mind the residual or error is minimum.

The below equation represents multiple linear regression in notation format.

$$y = b_0 + b_1 * x_1 + b_2 * x_2 + \varepsilon \quad \text{-(I)}$$

### B. Decision tree Regressor

Decision tree Regressor learning is a common predictive modeling technique used in machine learning. In decision tree, a model predicts by learning simple decision guidelines and splits the data into nodes with branches constructed on the basis of the best importance from the data. Decision tree regressor works on numeric data wherein the dependent variable are continuous numeric values. Decision trees are widely used machine learning algorithms due to their lucidity [5].

### C. Random Forest Regressor

Random Forest Regressor is an ensemble learning technique that works by training multiple decision trees and then outputs the average prediction (regression) of those trees. In general, Random forests shows better results compared to the decision tree. However, features can impact its performance [6] .

### D. XGBoost Regressor

XGBoost Regressor is also an ensemble learning technique but it focuses on efficient implementation of gradient boosting. Different decision trees are constructed and these trees are added orderly to the ensemble and fit to modify the errors and take corrective measures for the previous models.

## VI. EVALUATION METRICS & RESULTS

Multiple linear Regression model banks on linearity as an assumption and based on the results, the r squared value showcasing accuracy is merely 57%. The results are as below shown in Fig.5.

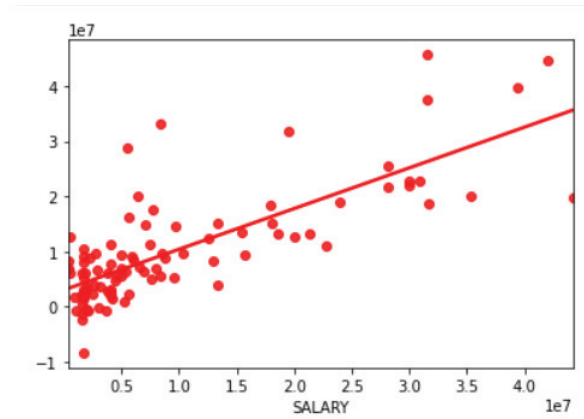


Fig. 5. Actual vs Predicted Salary

Variance Inflation Factors were calculated to check for autocorrelation. Fig.6 shows fantasy points and points have a strong correlation between them. This violates the assumption of Linear Regression. The other models used are Non-Linear and thus show better results.

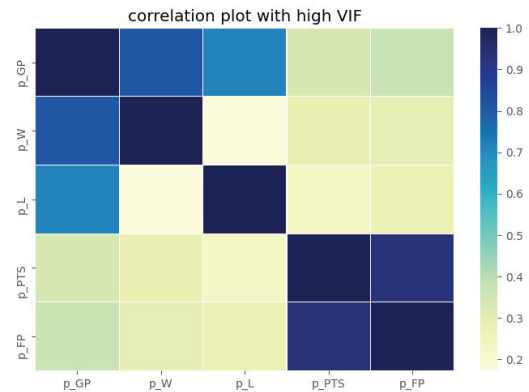


Fig. 6. Correlation within a Player's Performance Metric with High VIF

Decision Tree showed an improvement in the accuracy and the error. The accuracy recorded is 64% upgrading from 57%. A nonlinear link between pay and a few factors is seen in our sample. This explains why the decision tree regressor outperforms linear regression.

On the other XGB Regressor uses ensemble learning with boosting technique. While using XGBoost one tree is built at a time so that the data relating to the respective decision tree is taken into account and missing values are accounted for. This helps in working with gradient algorithms along with the decision tree algorithm which improves results. The R2 value turned out to be higher compared to the Regression tree i.e. 75.85%. The below image shows the relation amongst the predicted and actual values of the salary while using XGB Regressor

Finally, Random Forest leads to the accuracy score of R2 is 88.18%. The random forest algorithm prevents the model from overfitting by using multiple trees as shown in Table.2. The Root Mean Squared Error (RMSE) is the lowest. Each tree that is produced is based on a separate sample of data. While splitting, a separate sample of characteristics is picked at each node. Then, individual tree models make their predictions. Finally, these forecasts are averaged to get a single outcome. Moreover, while building the random forest model feature importance was checked to get an optimum model as shown in fig.7 and fig.8. Diagrammatic representation is as below.

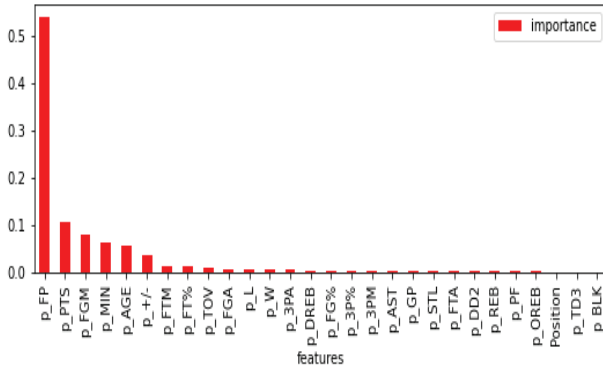


Fig. 7. Importance of Features in Predicting Player's Salary

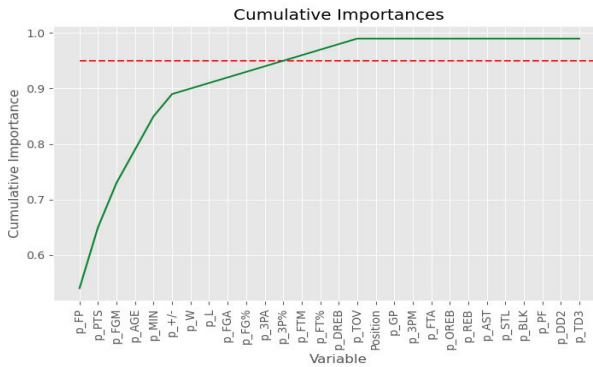


Fig. 8. Cumulative Importance of Features in Predicting Player's Salary

Table1 shows comparison of The R2 and Root Mean Squared Error across the models. The closer the value of R<sup>2</sup> to 1, better the accuracy of the model. Inversely, lower the

value of RMSE better the model as it shows smaller error values while predicting the model.

TABLE II. COMPARISON ACROSS MODELS

| Model Name                 | Split Ratio | RMSE         | R <sup>2</sup> |
|----------------------------|-------------|--------------|----------------|
| Multiple Linear Regression | 80:20       | 69,67,066.5  | 57.04%         |
| Decision Tree              | 80:20       | 57,71,902.35 | 64.4%          |
| XGB Regressor              | 80:20       | 47,54,531.87 | 75.85%         |
| Random Forest              | 80:20       | 33,25,775.78 | 88.18%         |

## VII. CONCLUSION

The results of our study specified the viability of a nonlinear model over a linear model while predicting the salary. There was also a breach of the Linear Regression assumption. Thus, the contributions of this work are not limited to pay prediction but also to the methodology utilised in contrast to the conventional strategy used to solve comparable problems (based on prior studies). Our model recorded the best available accuracy rate which depicts that the previous season's performance statistics are a good measure to predict the salary of the current year. The most important variable in predicting the salary was the fantasy points of a player as it showed the feature importance of 54 percent. Future study might focus on estimating wage for the next n years using a neural network if adequate data is available for deep learning. Furthermore, features such as the ratio of games played to losses, time spent at gym, marketing skills, and team playing ability can be utilized to improve salary prediction.

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